

Blood & Bone Marrow Histopathology - PRE-SAL Factoids

Week 7 SAL • summary-based draft • replace/add actual lab images after class

First question: peripheral blood or marrow? Then identify cells by **size + nucleus + cytoplasm + granules + central pallor**.

RBC morphology

Finding	Recognition	Attach mentally
Normal RBC	Uniform anucleate disc; central pallor ~1/3 diameter	Baseline size/color reference
Anisocytosis	Variable RBC sizes	RDW / mixed-size population
Microcytosis	Smaller RBCs; often increased central pallor	IDA / thal framework
Spherocyte	Dense round RBC, no central pallor	HS; warm AIHA can also do this
Sickle	Crescent / boat form with pointed ends	HbS polymerization
Reticulocytosis	Larger bluish-gray polychromatophilic RBC	Residual RNA = marrow response
Extra: schistocyte	Fragment / helmet cell	MAHA / mechanical shear
Extra: target cell	Bullseye appearance	Thalassemia / liver / Hb disorders

Spherocyte vs microcyte: no pallor = spherocyte; small + pallor = microcyte.

Five WBCs + band

Cell	Fast morphology	Main job / clue
Neutrophil	2-5 nuclear lobes, pale fine granules	Acute bacterial phagocyte
Band	U/horseshoe nucleus, not segmented	Immature neutrophil / left shift
Eosinophil	Bilobed + orange/red granules	Helminths, allergy; MBP
Basophil	Dark purple granules obscure nucleus	IgE/allergic mediator release
Lymphocyte	Round dark nucleus fills most of cell; thin blue rim	B/T/NK indistinguishable by routine morphology
Monocyte	Largest WBC ; folded/kidney nucleus; gray-blue vacuoles	Macrophage precursor / APC / phagocyte

Platelets

Tiny anucleate purple fragments from megakaryocytes. Alpha granules = vWF, fibrinogen, PDGF. Dense granules = ADP, Ca2+, serotonin. Primary-hemostasis cross-link: vWF-GpIb -> activation -> GpIIb/IIIa-fibrinogen aggregation.

Bone Marrow Speed Sheet

Peripheral blood vs marrow

Peripheral blood	Bone marrow aspirate / smear
Mostly mature anucleate RBCs	Many nucleated precursor cells
Scattered mature WBCs	Multiple granulocyte maturation stages
Tiny platelet fragments	Giant multilobulated megakaryocytes
No normal marrow spicules/adipose architecture	May see spicules, adipocytes, mixed precursor populations

Aspirate	Core biopsy
“What cells are here?” Cell morphology, differential, flow/cytogenetic material.	“How is the marrow built?” Architecture, overall cellularity, fibrosis, focal infiltrates.

Normal marrow landmarks

Megakaryocyte	Huge cell, multilobulated/polyloid nucleus; sheds platelets toward sinus
Erythroid island	Cluster of erythroid precursors around central macrophage
Myeloid precursors	Granular cytoplasm across maturation spectrum
Adipocytes	Clear round spaces / fat component
M:E ratio	Normal adult marrow roughly 2:1 to 4:1

Maturation sequences

Erythroid Proerythroblast -> basophilic -> polychromatophilic -> orthochromatic -> retic -> RBC. Trend: smaller; blue -> pink; nucleus condenses/extrudes.	Neutrophil Myeloblast -> promyelocyte -> myelocyte -> metamyelocyte -> band -> segmented. Trend: granules appear; nucleus indents then segments.
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Practical traps

- **Reticulocyte ≠ nucleated RBC:** retic is anucleate with residual RNA/polychromasia.
- **Band vs monocyte:** band = neutrophilic U-shaped nucleus; monocyte = larger folded/kidney nucleus + gray-blue cytoplasm.
- **T/B/NK cannot be separated** reliably by routine small-lymphocyte morphology.
- Blasts or nucleated RBCs in peripheral blood are not normal mature-blood findings and should make you stop and reassess.

Immediate marrow giveaway: giant megakaryocyte + many nucleated precursor stages.